Two quick inequalities

Demo paper for the TeX proof-check kit

Lemma 1. *For all real numbers a and b we have $a^2 + b^2 \geq 2ab$.*

Proof. We have $a^2 + b^2 - 2ab = (a - b)^2 \geq 0$. □

Lemma 2. *For every real number x with $x \leq 0$ or $x \geq 1$ we have $x^2 \geq x$.*

Proof. We have $x^2 - x = x(x - 1)$. If $x \leq 0$, both factors are nonpositive; if $x \geq 1$, both are nonnegative. In either case their product is nonnegative, and hence $x^2 \geq x$. □